GARUDA

AI-Powered Home Services Marketplace
for the People of Nepal

Technical Documentation
StartupNepal AI Hackathon 2026

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Team Desamorados

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Chapter 1

Executive Summary

Project Overview

Garuda is a full-stack, AI-powered home services marketplace built specifically for Nepal. It connects homeowners with verified service providers through transparent AI-assisted pricing, intelligent provider matching, and a multi-turn bilingual chatbot. A core differentiator is the **Women First** system, which actively prioritises and surfaces female service providers.

The platform was designed and built by **Anamol Khadka** and **Abishek Parajuli** as the submission for the StartupNepal AI Hackathon 2026 under the team name *Desamorados*.

1.1 Key Metrics at a Glance

10	7+	20	3
Service Categories	AI Endpoints	req/min rate limit	User Roles

Chapter 2

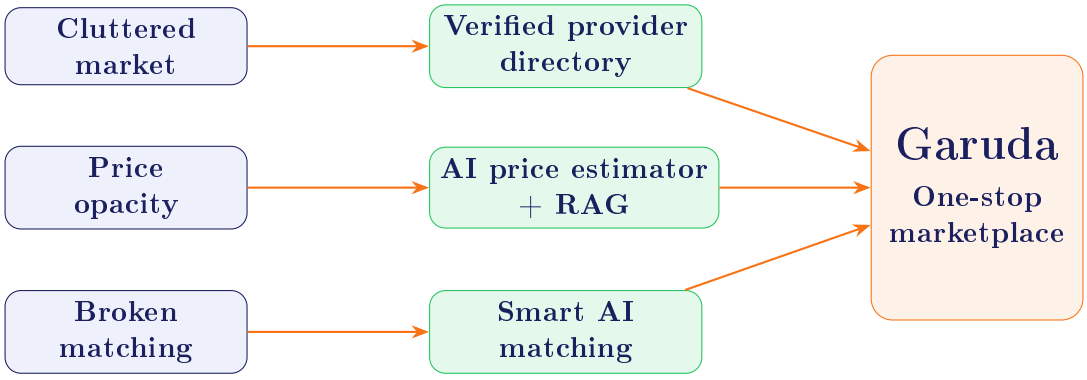
Problem Statement & Solution

2.1 Problem Landscape

Nepal's home services sector suffers from three interconnected problems:

1. **Cluttered & informal market** — homeowners rely on word-of-mouth or random street contacts; no platform aggregates supply.
2. **Price opacity & quality uncertainty** — no transparent estimates or service guarantees exist.
3. **Broken matching** — workers wait at *chowks* and depend on middlemen who take high commissions.

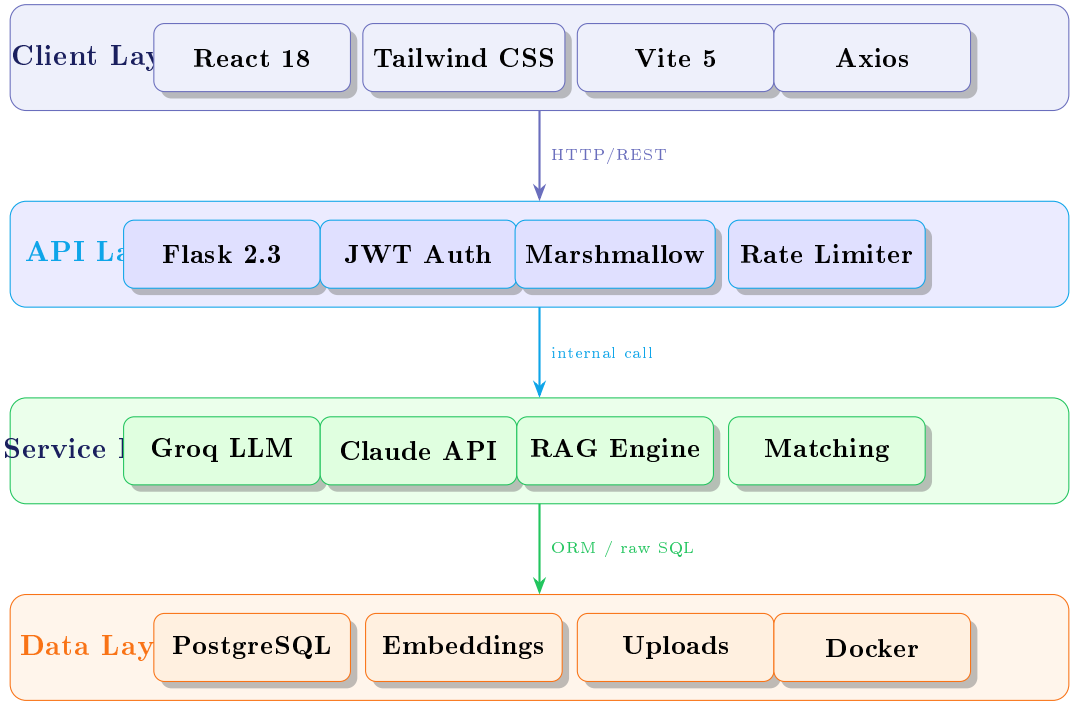
2.2 Solution Map



Chapter 3

System Architecture

3.1 High-Level Architecture Diagram



3.2 Technology Stack

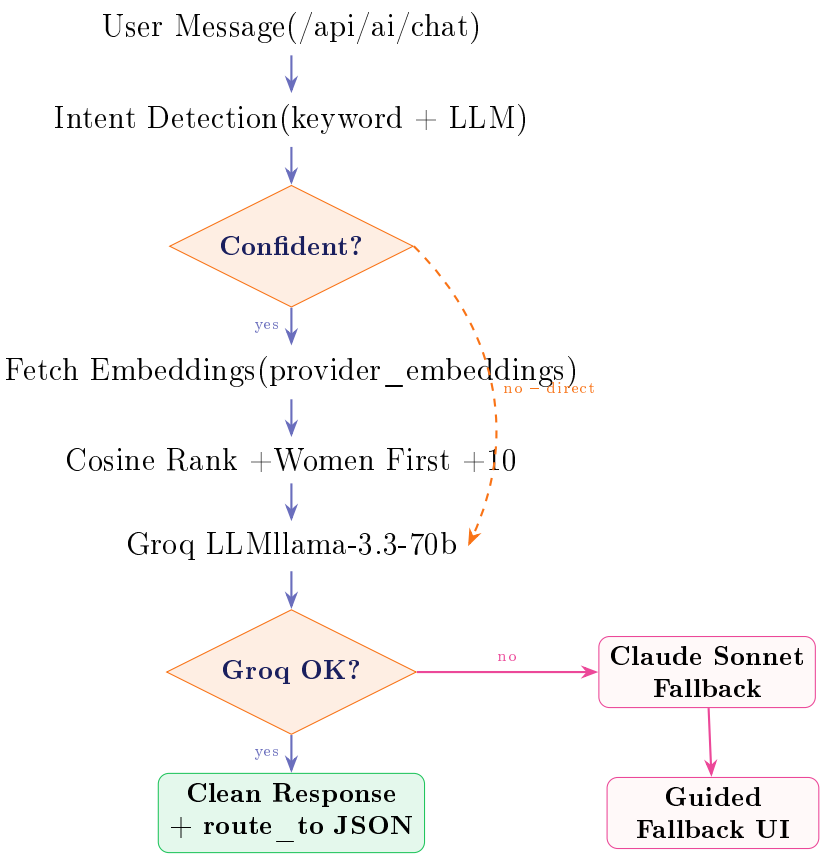
Full Stack Overview

Layer	Technology	Notes
Frontend	React 18.2 + Vite 5	Mobile-first SPA
Styling	Tailwind CSS 3	Custom Garuda theme
HTTP Client	Axios	JWT auto-attach via client.js
Backend	Flask 2.3.5	REST API, Gunicorn in prod
Database	PostgreSQL 15 + SQLAlchemy	Flask-Migrate migrations
Auth	Flask-JWT-Extended	24 h access tokens
Primary LLM	Groq llama-3.3-70b-versatile	Low-latency inference
Fallback LLM	Claude Sonnet 4	claude-sonnet-4-20250514
Vectors	pgvector cosine similarity	provider_embeddings table
DevOps	Docker + Docker Compose	Bridge network, health checks

Chapter 4

AI Architecture

4.1 LLM + RAG Pipeline Flowchart



4.2 AI Endpoints Reference

Endpoint	Auth	Description
/api/ai/chat	None	Multi-turn bilingual chatbot (Nepali/English)
/api/ai/match-providers	None	Ranked provider list with Women First bonus
/api/ai/estimate-price	None	RAG-powered price estimation
/api/ai/extract-booking	None	Parse booking intent from text or image
/api/ai/analyze-image	None	Detect issue from photo, suggest service
/api/ai/summarize-reviews	JWT	2–3 sentence provider review summary
/api/ai/trust-score	JWT	0–100 trust score with badge assignment

*All AI endpoints are rate-limited to **20 requests per minute per user**.*

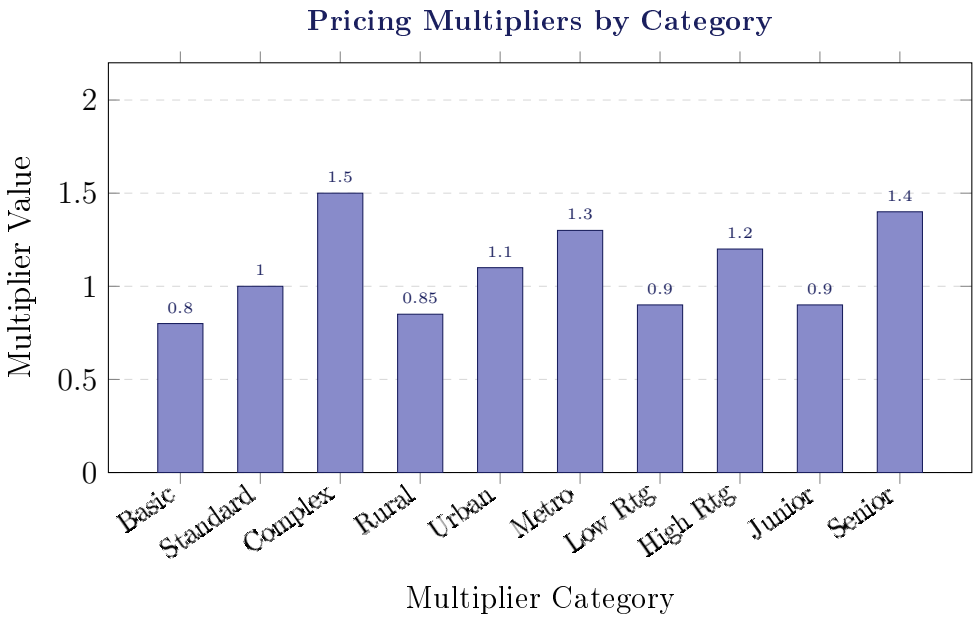
4.3 Dynamic Cost Calculator Formula

Pricing Formula (DynamicCostCalculator.jsx)

$$\text{Final Price} = \underbrace{B}_{\text{base}} \times \underbrace{C_x}_{\text{complexity}} \times \underbrace{L_x}_{\text{location}} \times \underbrace{R_x}_{\text{rating}} \times \underbrace{E_x}_{\text{experience}} + \underbrace{d \times 10}_{\text{travel (Rs/km)}}$$

where d is the Haversine distance (km) between provider and customer. Travel cost is **Rs 10/km**, computed client-side.

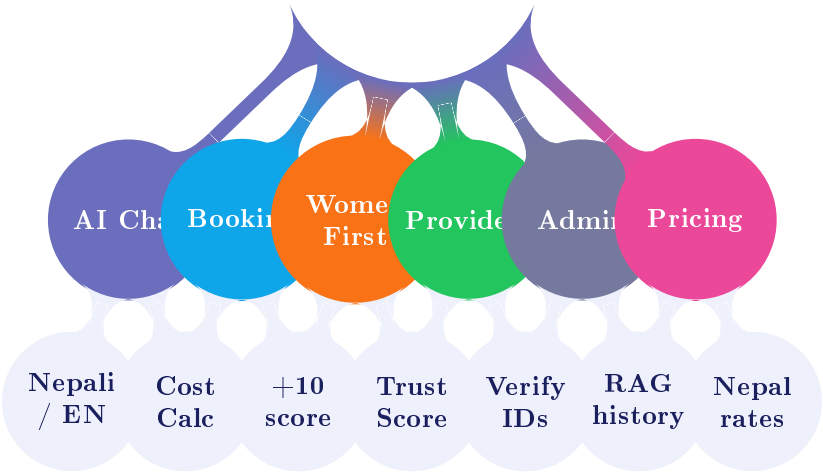
4.4 Pricing Multiplier Bar Chart



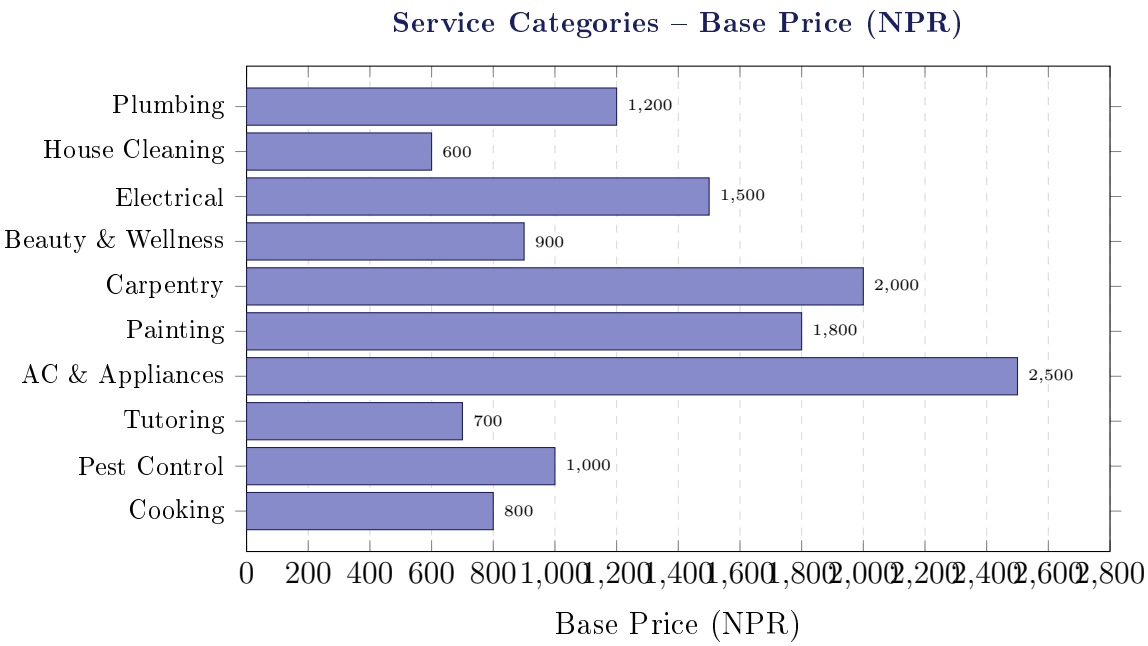
Chapter 5

Features & Functionality

5.1 Core Feature Mind Map



5.2 Service Categories Base Price Chart



Chapter 6

Women First System

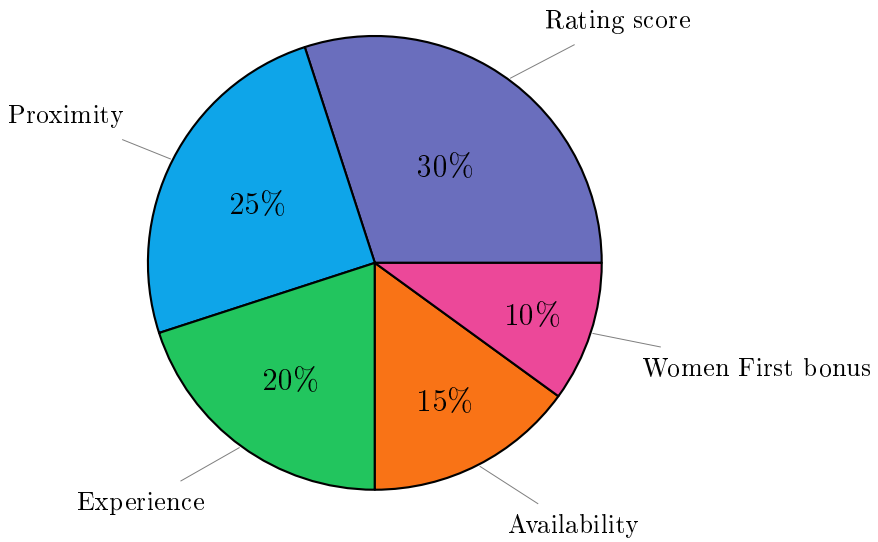
The Women First system is a deliberate, multi-layered approach to surface and prioritise female service providers on the platform, directly addressing the gender participation gap in Nepal's informal labour market.

6.1 Implementation Layers

Visual layer	**UI: Women First badge on every female provider card**
Filter layer	**UX: Toggle filter on Services page**
Algorithm layer	**Algorithm: +10 bonus in AI match score**
API layer	**API: women_first=true param on /api/providers**
Data layer	**DB: is_female boolean + ORDER BY is_female DESC**

Women Fi Stack

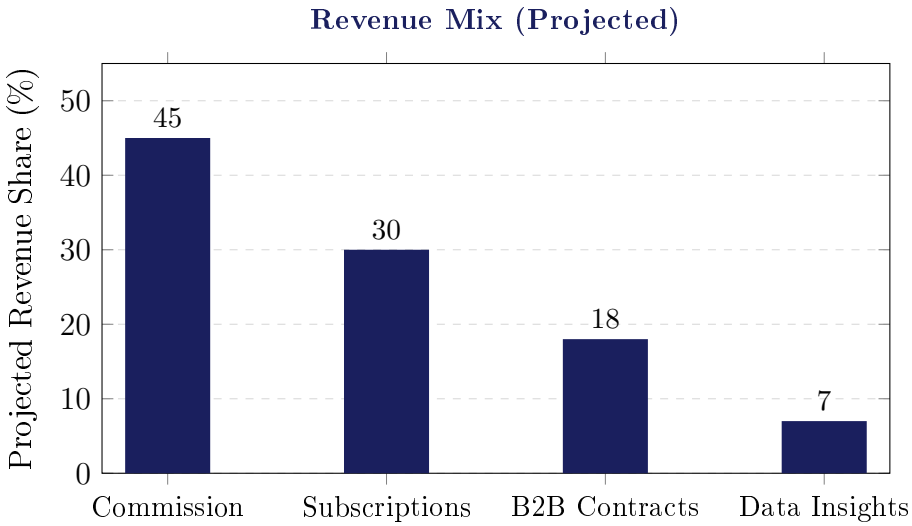
6.2 Provider Matching Score Breakdown



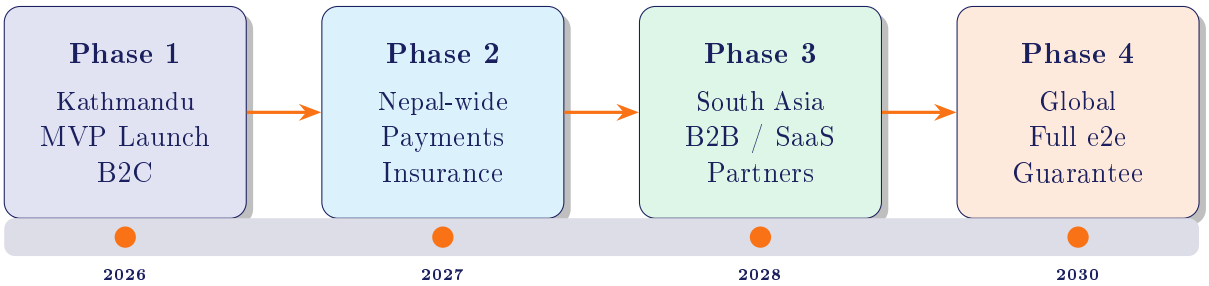
Chapter 7

Revenue Model & Scaling

7.1 Revenue Streams Bar Chart



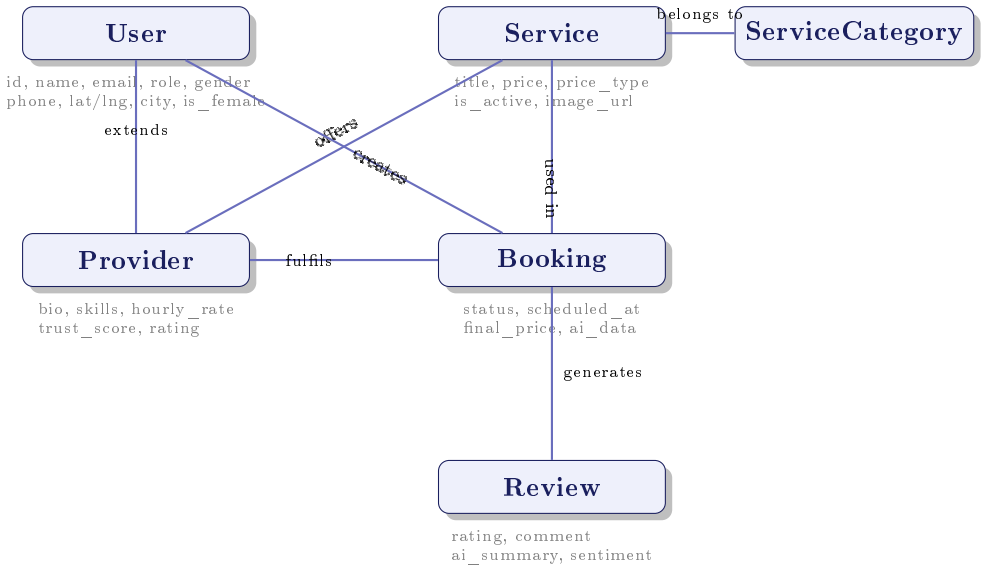
7.2 Scaling Roadmap



Chapter 8

Database Schema

8.1 Entity-Relationship Diagram



8.2 Database Models Summary

Core Models

Model	Key Fields
User	id, name, email, role, gender, is_female, lat/lng, city
Provider	bio, skills, years_experience, rating, trust_score, trust_badge
ServiceCategory	id, name, name_np (Nepali), icon, base_price
Service	provider_id, category_id, title, price, price_type
Booking	customer_id, provider_id, status, ai_extracted_data, final_price
Review	booking_id, rating, comment, ai_summary, sentiment

Chapter 9

API Reference

9.1 Authentication Endpoints

Method	Endpoint	Auth	Purpose
POST	/api/auth/register	None	Register customer or provider
POST	/api/auth/login	None	Login, receive JWT tokens
POST	/api/auth/refresh	Refresh Token	Obtain new access token
GET	/api/auth/me	JWT	Get current user profile

9.2 Provider & Service Endpoints

Method	Endpoint	Auth	Purpose
GET	/api/providers	None	List providers (filter: city, rating, women_first)
GET	/api/providers/:id	None	Provider profile and reviews
PUT	/api/providers/:id	JWT	Update profile
POST	/api/providers/:id/verify	Admin	Verify identity
GET	/api/services	None	List services (filterable)
POST	/api/services	JWT	Create service

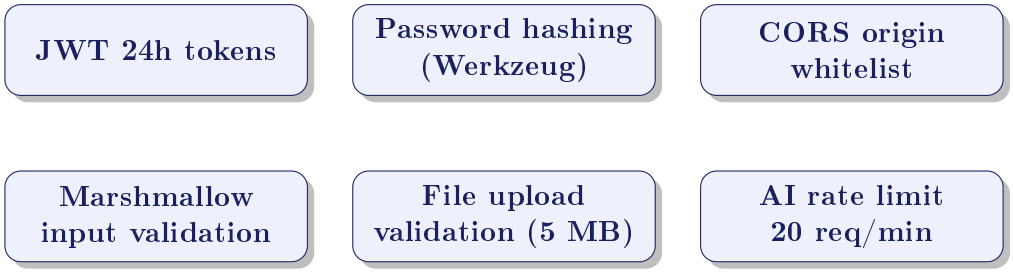
9.3 Booking & Review Endpoints

Method	Endpoint	Auth	Purpose
POST	/api/bookings	JWT	Create booking
GET	/api/bookings/my	JWT	Customer booking list
GET	/api/bookings/assigned	JWT	Provider booking list
PUT	/api/bookings/:id/status	JWT	Update booking status
POST	/api/reviews	JWT	Submit review
GET	/api/reviews/provider/:id	None	Get provider reviews

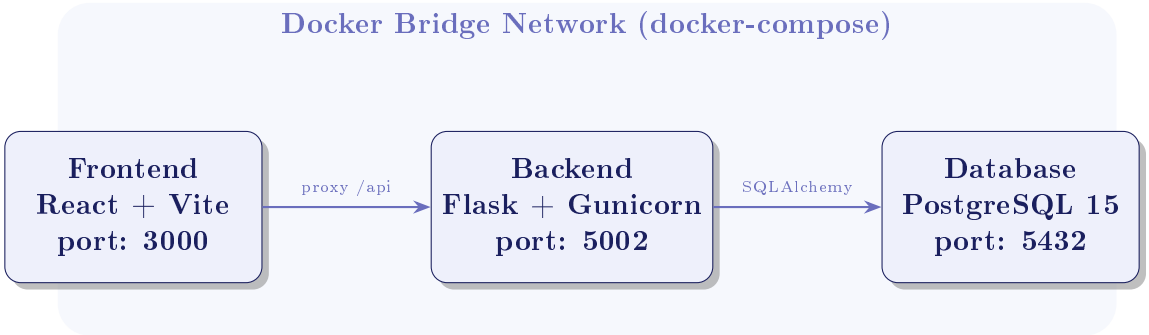
Chapter 10

Security & Infrastructure

10.1 Security Measures



10.2 Docker Service Architecture



Chapter 11

Future Work & Limitations

11.1 Current MVP Limitations

Limitation	Planned Resolution
Payment integration mocked	Integrate eSewa / Khalti
Google Maps optional for MVP	Full integration in Phase 2
Local image storage	Migrate to AWS S3
Notifications UI only	Backend + SMS/Email integration

11.2 Feature Priority Radar

Chapter 12

Setup Guide

12.1 Prerequisites

- Docker and Docker Compose (macOS, Linux, or Windows WSL2)
- Anthropic API key (Claude Sonnet 4 fallback LLM)
- Groq API key (primary LLM)
- Optional: Google Maps API key

12.2 Quick Start

```
# 1. Clone the repository
git clone https://github.com/StartupNepal-AI-Hackathon-2026/Desamorados
cd Desamorados

# 2. Create environment file
cp .env.example .env
# Edit .env: set ANTHROPIC_API_KEY, JWT_SECRET_KEY, GROQ_API_KEY

# 3. Launch all services
docker-compose up -d

# 4. Seed sample data
docker-compose exec backend python seed_db.py

# Frontend: http://localhost:3000
# Backend: http://localhost:5002
```

12.3 Environment Variables Reference

.env variables

Variable	Purpose
ANTHROPIC_API_KEY	Claude API key (fallback LLM)
JWT_SECRET_KEY	Token signing secret
GOOGLE_MAPS_API_	Distance and map features
DATABASE_URL	PostgreSQL connection string
FLASK_ENV	development / production
BACKEND_PORT	Default 5002
VITE_API_URL	Frontend proxy target

GARUDA

Built to empower Nepal's service economy,

one booking at a time.

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Team Desamorados — StartupNepal AI Hackathon 2026